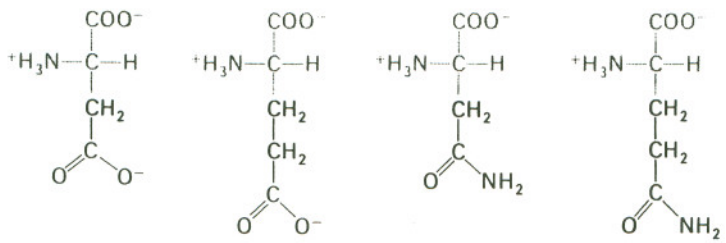
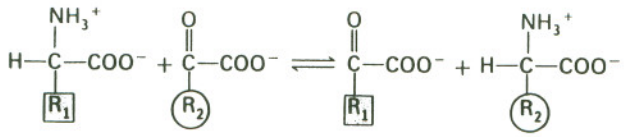
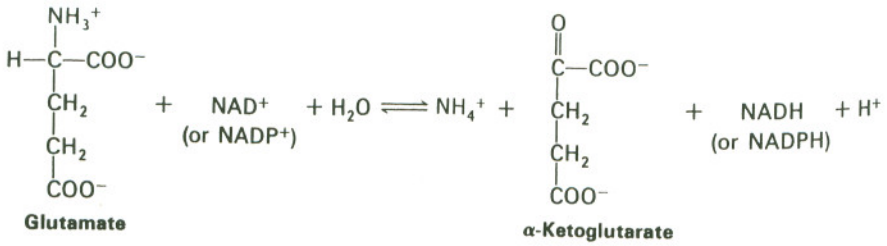
BIOCHEMISTRY CUME
NOVEMBER 16, 2001

A. (40) Write a complete enzymatic pathway for the synthesis of aspartate in a liver cell suspension, using glutamate as the <u>sole</u> source of both carbon and nitrogen. You may assume that other metabolites are present in small catalytic amounts.

B. (30) Write a complete balanced equation for the conversion of glutamate to aspartate in the liver cell under aerobic conditions, giving the number of CO_2 generated, O_2 consumed, and ATP produced.

C. (30) Determine which carbon atom(s) of aspartate will <u>first</u> become labeled if glutamate labeled with C-14 at carbon number 3 is used?

Glutamate: $^{-}OOC\text{-}CHNH_3^{+}\text{-}^{*}CH_2\text{-}CH_2\text{-}COO^{-}$